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AN2DL - First Challenge Report
P.E.K.K.A.tori

Tommaso Alongi, Federico Carugati, Francesco Paolini, Alessandro Romani

tommasoalongipolimi, fedepolitecnico, francescopaolini, aleromani4

306455, 303715, 300560, 303256

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1 Introduction

The following project focuses on training and building a *Convolutional Neural Network* (CNN) model to analyze different types of microscopic tissue morphology and predict the correct molecular subtype.

The core problem is the *multi-class classification* of breast cancer molecular subtypes: **Luminal A, Luminal B, HER2(+), Triple Negative**.

1.1 Project's Goals

The primary objectives of this project are to develop a **robust histological image classification model** capable of high generalization and to achieve a high F1-score on unseen test data, ensuring a balanced trade-off between precision and recall.

1.2 The Team's Approach

The workflow consisted of various main phases: data loading; dataset cleaning and preprocessing; development of different architectures, including both hand-made CNNs and various pretrained models taken from PyTorch library (Transfer Learning); performance refinement via hyperparameters tuning and fine tuning of pretrained models through layers unfreezing.

2 Problem Analysis

2.1 Dataset Characteristics Overview

The dataset consists of image-mask pairs derived from low-magnification Whole Slide Images (WSI) of human tissue. Each image is meticulously annotated with a **binary mask** to delineate regions highly predictive of **pathological tissue**. Crucially, every annotated pair is further classified by its corresponding **molecular subtype**. A pronounced imbalance among the four classes was observed, leading to several considerations later discussed in the subsequent sections.

2.2 Main Challenges

Our initial challenge involved **data curation and quality control**. The raw dataset exhibited significant anomalies and artifacts, including the presence of maliciously introduced features such as highly saturated color spots and extraneous pictorial content (e.g., images of the character Shrek). This necessitated the implementation of a rigorous **data cleaning protocol**.

Following the successful remediation of these data quality issues, several crucial methodological decisions arose. First, we addressed the **mask integration strategy**: determining the optimal method for incorporating the binary masks into our models, given their critical informational value for

identifying pathological regions.

Second, we tackled **resolution management**: developing an effective approach to process the high-resolution input images to prevent a substantial reduction in spatial information before feeding them into the model architecture.

3 Method

Our data cleaning procedure began by implementing an algorithm to compute the **cosine similarity** between the images in the dataset and the known set of maliciously introduced images. This initial algorithmic step served as a preliminary high-throughput data filtering mechanism, which was subsequently followed by a final manual curation pass to ensure a robust and clean dataset for analysis.

To address the challenge of high-resolution inputs and mitigate potential information loss during downsampling, we adopted a **patch-based processing strategy** [1]. Our model was trained on representative patches extracted from the Whole Slide Images (WSI), with the definitive image-level prediction obtained by aggregating the patch-level results using a **majority voting system**.

To prevent the model from processing low-information-value patches, a crucial preprocessing step involved **filtering**: specifically, we discarded any patch where the percentage of white pixels (representing the pathological region) within the associated binary mask fell below a predetermined **threshold**. Furthermore, in later experiments, we implemented an artifact mitigation strategy to handle images containing color spots by replacing these artifacts with white color; this allowed us to retain the informative value of the image while simultaneously filtering patches that contained an excessive number of bright pixels.

To manage class imbalance within the training set, we employed a **weighted cross-entropy loss function**, where the weights were calculated based on the class distribution in the training split, adjusted by an introduced smoothing term.

Finally, **Image Augmentation (IA)** [2] [4] was extensively utilized throughout the training phases and applied during the evaluation stage through **Test Time Augmentation (TTA)** to enhance model robustness and generalization capabilities.

4 Experiments

This section presents the experimental evaluation, reporting results from various baselines trials to progressively enhanced models incorporating image patching, weighted loss and other important improvements.

All experiments were conducted with fixed random seeds for reproducibility and the macro-averaged F1-score was used as the primary metric.

4.1 Baseline Models

Initial experiments were performed by resizing the dataset images to a resolution of 560×560 pixels. The first trained model was a vanilla convolutional neural network (CNN) architecture. As anticipated, this configuration resulted in limited generalization capability. Subsequently, model evaluation was extended by leveraging transfer learning with several state-of-the-art pre-trained architectures available in the PyTorch framework. The investigated models included EfficientNet, Swin Transformer—widely regarded as one of the most effective architectures for histopathological image classification alongside ConvNeXt—and ResNet. The performance of these models on the validation set is reported in Table 1.

Table 1: Performance Overview over the validation set of baseline experiments

Model Variant	Accuracy	F1
Vanilla CNN	0.3563	0.1872
EfficientNetB0	0.3563	0.3266
ResNet	0.4138	0.3847
SwinTransformer	0.4023	0.3784

As reported in Table 1, the baseline models exhibit overall poor performance. An analysis of the corresponding confusion matrices reveals that this behavior is primarily driven by an incorrect yet predominant prediction of the Luminal B subtype. Among the evaluated models, the ResNet appears to be the most stable, although the observed improvements in F1-score remain modest.

4.2 Further implementations

To address class imbalance, the limited size of the dataset—further reduced by data cleaning—and

to improve the overall performance of the models, several complementary strategies were investigated. Significant performance gains were achieved by introducing an image patching strategy combined with the use of image masks, as described in the Method section. Lastly, a weighted loss function was adopted. These experiments were conducted using a **ConvNeXt** architecture.

Since image patches often encode heterogeneous information and only a limited subset of regions is typically relevant for cancer diagnosis, we explored alternative modeling strategies beyond standard patch-based approaches. Among the additional methods investigated, we adopted the *Clustering-constrained Attention Multiple Instance Learning* (**CLAM**) framework [3]. This approach extends the MIL paradigm by operating on feature representations extracted from individual patches and employs an attention-based mechanism to aggregate instance-level information into a bag-level prediction. At the same time, CLAM introduces clustering constraints that explicitly encourage the separation of informative instances from non-informative ones, making it particularly well suited for scenarios in which discriminative regions are sparse.

Furthermore, an **ensemble strategy** was evaluated by combining predictions from multiple independently trained classifiers— adopting different architectures and training configurations—using majority voting, with the aim of enhancing robustness and overall predictive performance.

We also experimented with models accepting **four input channels**, with one channel explicitly containing the binary mask information, although these multi-channel models did not demonstrate superior performance.

Table 2: Performance Overview over the validation set after improvements

Model Variant	F1
ConvNeXt	0.4037
CLAM	0.7908
FourChannels	0.3805

4.3 Final Model Selection

The Ensemble Model achieved the best test-set performance, reaching an **F1-score (test)** of **0.4211** and was selected as the final architecture for deployment.

5 Results

The overall best performance on the test set was obtained by **ensembling** the three preceding top-performing models using a **majority voting system**. These constituent models were all based on the ConvNeXt and Phikon architecture, each trained with a distinct combination of patch size and stride. Although the **CLAM** model demonstrated outstanding performance on the validation set, this level of efficacy was not replicated on the test data. Further details and a discussion of this observed **anomalous generalization gap** are provided in the subsequent section.

6 Discussion

As shown in Table 2, the high F1-score of 0.7908 achieved on the validation set using the CLAM architecture should be interpreted with caution. While this result may suggest strong performance, it does not guarantee comparable behavior on the test set. The observed discrepancy may indicate issues such as overfitting or potential inconsistencies in the evaluation or testing pipeline, and therefore warrants further investigation.

7 Conclusions

It can be concluded that the utilization of **patches**—which allows for the usage of **high-resolution images**—combined with a **majority voting-based system** and the introduction of a **weighted cross-entropy loss**, yields notable improvements in model performance. Furthermore, a correct implementation of **masking** can refine the accurate execution of the classification task.

Future enhancements and experiments could involve further exploration of **CLAM** (Clustering-constrained Attention Multiple Instance Learning) architectures, as well as a more meticulous tuning of **hyperparameters** to determine the optimal combination of patch size and stride.

References

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